

Globular Homotopy Theory of Spaces

Bar Roytman

1 Introduction

Theorem 1: For any coherator, the Grothendieck–Quillen model structure on the category of models over the coherator exists. Furthermore, every topological realization of a coherator induces a Quillen equivalence between the category of spaces with the Quillen model structure and models of a coherator with the Grothendieck–Quillen model structure. Homotopies between topological realization give rise to natural weak equivalences between the corresponding Quillen adjunctions.

Theorem 2: Every cellular extension of cellular globular extensions that are equipped with standard model structures such that the right adjoint preserves cofibrations and acyclic cofibrations and unit applied to cell complexes is a weak equivalence induces a Quillen equivalence. If the source of the cellular extension receives a faithful globular functor from the globular completion with image in the subcategory of monomorphisms, the right adjoint preserves cofibrations and the unit is a weak equivalence under the additional hypothesis that the unit is a weak equivalence on globes. In particular, every cellular extension of models over a coherator with Grothendieck–Quillen model structures is a Quillen equivalence.

Theorem 3: Every Ara coherator is a coherator.

2 Cellular Globular Extensions

We name every map in a cellular globular extension and compute the category explicitly.

Let $\mathbf{G}$ be the directed graph with vertices $\mathbb{N}$ and edges $\sigma^n, \tau^n: n \rightarrow n+1$

Let G be the free category on $\mathbf{G}$.

The category of globes $\mathbb{G}$ is the the quotient category of G modulo the relations relations $\sigma^{n+1} \circ \sigma^n = \tau^{n+1} \circ \sigma^n$ and $\sigma^{n+1} \circ \tau^n = \tau^{n+1} \circ \tau^n$ with quotient functor $\pi: G \rightarrow \mathbb{G}$.

We define for $n, m \in \mathbb{N}$ with $n > m$ the morphisms $\sigma_n^m, \tau_n^m: m \rightarrow n$ of $\mathbb{G}$ inductively by $\sigma_{m+1}^m = \sigma^m$, $\tau_{m+1}^m = \tau^m$, and for all $k > n$, $\sigma_{k+1}^m = \sigma^k \circ \sigma_k^m$ and $\tau_{k+1}^m = \sigma^k \circ \tau_k^m$.

Every morphism in $\mathbb{G}$ is exactly one of id_n for some necessarily unique $n \in \mathbb{N}$, σ_n^m for some necessarily unique $m, n \in \mathbb{N}$ with $m < n$, and τ_n^m for some necessarily unique $m, n \in \mathbb{N}$ with $m < n$.

Proof: Step 1: If $f: k \rightarrow \ell$ is a morphism of $\mathbb{G}$ of length 0, then $\pi(f) = \text{id}_k$. Step 2: If $f: k \rightarrow \ell$ is a morphism of $\mathbb{G}$ of length 1, then $\ell = k + 1$ and $\pi(f) = \sigma_\ell^k$ or $\pi(f) = \tau_\ell^k$. Step 3: Suppose every morphism of $g: a \rightarrow b$ in $\mathbb{G}$ of length i satisfies $\pi(g) = \sigma_b^a$ or $\pi(g) = \tau_b^a$. Then, for every morphism $f: k \rightarrow \ell$ in $\mathbb{G}$ of length $i + 1$ satisfies $\pi(f) = \sigma_\ell^k$ or $\pi(f) = \tau_\ell^k$. Step 4: If $f, g: k \rightarrow \ell$ are morphisms of $\mathbb{G}$ and $\pi(f) = \pi(g)$, then f and g have the same length. Step 5: Suppose $f: k + 1 \rightarrow \ell$ is a morphism of $\mathbb{G}$. Then every $g: k \rightarrow \ell$ such that $\pi(g) = \pi(f \circ \sigma^k)$ is of the form $g = g' \circ \sigma^k$ for some $g': k + 1 \rightarrow \ell$. Similarly, every $h: k \rightarrow \ell$ such that $\pi(h) = \pi(f \circ \tau^k)$ is of the form $h = h' \circ \sigma^k$ for some $h': k + 1 \rightarrow \ell$. Step 6: Put together these facts.

A table $T = (k, (a_i)_{i \in [k]}, (b_j)_{j \in [k-1]})$ consists of a natural number k called the width, a sequence $(a_i)_{i \in [k]}$ of natural numbers called the generator dimensions, and a sequence $(b_j)_{j \in [k-1]}$ of natural numbers called the gluing dimensions, satisfying $a_j > b_j < a_{j+1}$ for all $j \in [k-1]$. The set of tables is denoted Table .

Let $\mathcal{Z}_k$ be the category specified as follows. The object set of $\mathcal{Z}_k$ is $[k] \amalg [k-1]$ with α_i denoting the image of $i \in [k]$ and β_j denoting the image of $j \in [k-1]$ in $[k] \amalg [k-1]$ under the coprojections. The category $\mathcal{Z}_k$ is the free category on the graph with edges $\ell_j: \beta_j \rightarrow \alpha_j$ and $r_j: \beta_j \rightarrow \alpha_{j+1}$ for all $j \in [k-1]$.

The length 0 morphisms of $\mathcal{Z}_k$ are identities, the length 1 morphisms are the edges of the graph, and since every edge has target in some α_i and source in some β_j , there is no target equal to any source and there are no length 2 morphisms. Consequently there are no length $2 + m$ morphisms for any $m \in \mathbb{N}$. The classification easily follows.

Every morphism of $\mathcal{Z}_k$ is precisely one of ℓ_j , r_j , id_{α_i} , or id_{β_j} for precisely one index $i \in [k]$ or $j \in [k-1]$ as appropriate.

For each table $T = (k, (a_i)_{i \in [k]}, (b_j)_{j \in [k-1]})$, the functor $\mathbf{Z}_T: \mathcal{Z}_k \rightarrow \mathbb{G}$ is the unique functor with $\mathbf{Z}_T(\alpha_i) = a_i$ for all $i \in [k]$, $\mathbf{Z}_T(\beta_j) = b_j$ for all $j \in [k-1]$, $\mathbf{Z}_T(\ell_j) = \sigma_{a_j}^{b_j}$, and $\mathbf{Z}_T(r_j) = \tau_{a_{j+1}}^{b_j}$.

A globular extension is a category $\mathcal{C}$ equipped with a functor $F: \mathbb{G} \rightarrow \mathcal{C}$ such that for every table T , $F \circ \mathbf{Z}_T$ has a colimit.

A globular functor $H: (\mathcal{C}, F) \rightarrow (\mathcal{C}', F')$ between globular extensions

$(\mathcal{C}, F), (\mathcal{C}', F')$ is a functor $H: \mathcal{C} \rightarrow \mathcal{C}'$ such that $H \circ F = F'$ and H preserves the colimits of $F \circ \mathbf{Z}_T$ for each table T .

The notions of natural transformations and natural isomorphisms of between globular functors are inherited from the underlying functors.

A globular extension $(\mathcal{C}, F)$ is a globular completion if for any globular extension $(\mathcal{C}', F')$, there exists a globular functor unique up to natural isomorphism $H: (\mathcal{C}, F) \rightarrow (\mathcal{C}', F')$.

While a globular completion is completely characterized by its universal property and examples of globular completions are plenty in the literature, we will still benefit from giving our own. While it is not a canonical description by any means, its use will reduce the logical complexity of the arguments to follow.

For a category $\mathcal{C}$, let $\mathcal{C}^\triangleright$ denote the category obtained by adjoining a terminal object $*$ to $\mathcal{C}$ via morphisms $!_a: a \rightarrow *$ for every object $a \in \mathcal{C}$.

The subglobe category $\mathcal{C}_0^p$ is defined as the pushout of categories $\mathbb{G} \leftarrow \coprod_{T \in \text{Table}} \mathcal{Z}_k \rightarrow \coprod_{T \in \text{Table}} \mathcal{Z}_k^\triangleright$.

The objects of $\mathcal{C}_0^p$ are identified with $\mathbb{N} \amalg \text{Table}$ with $\mathbb{N}$ corresponding to the objects of $\mathcal{G}$ and each $T \in \text{Table}$ corresponding to the object $*$ from the T -summand of $\coprod_{T \in \text{Table}} \mathcal{Z}_k^\triangleright$.

3 Models over Cellular Globular Extensions

We name every element of the target of relative cell complexes.

We prove relative cell complexes are closed under the forgetful functor of a cellular extension of globular extension.

We prove that for a cellular extension of globular extensions such that the source receives a faithful globular functor from the globular completion whose image is contained in the subcategory of monomorphisms, the right adjoint of the induced adjunction between categories of models preserves cofibrations and the unit is a weak equivalence on cellular objects whenever it is a weak equivalence on representables.

We prove that every cellular extension of cellular globular extensions that are equipped with standard model structures such that the right adjoint preserves cofibrations and acyclic cofibrations and unit applied to cell complexes is a weak equivalence induces a Quillen equivalence.

4 Constructions in Coherators and Models

We construct various important morphisms in a admissibly filled cellular extension with chosen admissible fillings.

We construct the first frustum cylinder functor and prove its basic properties.

We construct the barycentric subdivison functor and prove its basic properties.

In this section we should explicitly name the fillings we will use (Notation subject to change):

i_k fills (ι_k, ι_k) where $\iota_k: G^k \rightarrow G^k$ is the canonical isomorphism
 $c_{n,p}^m$ for $n > m < p$ fills $(\tau_T^{n,0}, \sigma_T^{n,1})$ when $n = p$, $(\langle \text{id}_n, \tau^q \rangle \circ c_{n,q}^m, \langle \text{id}_n, \sigma^q \rangle \circ c_{n,q}^m)$ when $n \leq q < q+1 = p$, and $(\langle \tau^r, \text{id}_p \rangle \circ c_{r,q}^m, \langle \sigma^r, \text{id}_p \rangle \circ c_{r,q}^m)$ when $n = r+1 > r \geq p$.

ρ_k fills $(\sigma_{\mathbf{k}+1,0}^k, \tau_{\mathbf{k}+1,0}^k)$.

We define CW-complexes B^m as follows. The cells are $e_{f,a}: G^{b+\delta_{a,1}} \rightarrow B^m$ where $f: b \rightarrow m$ and $a \in \{0, 1, 2\}$. When $a \in \{0, 2\}$ and $b \geq 1$ we have $(\sigma_b^{b-1})^* e_{f,a} = e_{f \circ \sigma_b^{b-1}, a}$ and $(\tau_b^{b-1})^* e_{f,a} = e_{f \circ \tau_b^{b-1}, a}$. We have $(\sigma^b)^* e_{f,1} = x_{f,a,b}$ and $(\tau^b)^* e_{f,1} = x_{f,b}$ where $x_{f,c}$ for all $j \in [b]$ are defined as follows. We have $x_{f,0} = e_{f,0}$, $x_{f,j+1} = (c_{b,j+1}^j)^* \langle (c_{j+1,b}^j)^* \langle e_{f \circ \tau_b^j, 1}, x_{f,j} \rangle, (\rho_j)^* \langle e_{f \circ \sigma_b^j, 1} \rangle \rangle$ for all $j \in [b-1]$.

The 0-skeleton is trivially well-defined. One checks the 1-skeleton is well defined by inspection that the boundary of 1-cells is comprised of 0-cells. Suppose the $i, i+1$ -skeleta are well-defined for some $i \in \mathbb{N}$ and that $x_{f,j}$ for $f: b \rightarrow m$ is a dimension b element of the $b+1$ -skeleton for all $b \leq i$ and that they satisfy for $j \leq a < b \leq i$, $(\sigma_b^a)^* x_{f,j} = x_{f \circ \sigma_b^a, j}$ and $(\tau_b^a)^* x_{f,j} = x_{f \circ \tau_b^a, j}$. Then, for $f: i+2 \rightarrow m$, and $a \in \{0, 2\}$, $e_{f \circ \sigma_{i+2}^{(i+2)-1}, a}$ and $e_{f \circ \tau_{i+2}^{(i+2)-1}, a}$ are dimension $i+1$ elements in the $(i+1)$ -skeleton with $(\sigma_{i+1}^i)^* e_{f \circ \sigma_{i+2}^{(i+2)-1}, a} = e_{f \circ \sigma_{i+2}^{i+1} \circ e_{i+1}^i, a} = e_{f \circ \tau_{i+2}^{i+1} \circ e_{i+1}^i, a} = (\sigma_{i+1}^i)^* e_{f \circ \tau_{i+2}^{(i+2)-1}, a}$. For $f: i+1 \rightarrow m$ and $j \in [b]$ we check that $x_{f,j}$ is a well-defined dimension $i+1$ element in the $(i+1)$ -skeleton by induction on j . The $j=0$ case is by inspection. Assume $x_{f,j}$ is a dimension $i+1$ element of the $(i+1)$ -skeleton. Then $(c_{j+1,i+1}^j)^* \langle e_{f \circ \tau_{i+1}^j, 1}, x_{f,j} \rangle$ is a dimension $i+1$ element of the $(i+1)$ -skeleton since $(\sigma_{j+1}^j)^* e_{f \circ \tau_{i+1}^j, 1} = x_{f \circ \tau_{i+1}^j, j} = (\tau_{i+1}^j)^* x_{f,j}$. By inspection, we see $(\rho_j)^* \langle e_{f \circ \sigma_{i+1}^j, 1} \rangle$ is a dimension j element of the $(i+1)$ -skeleton. Therefore, the existence of $x_{f,j+1}$ as a dimension $i+1$ element of the $(i+1)$ -skeleton follows from

$$\begin{aligned}
(\sigma_{i+1}^j)^*(c_{j+1,i+1}^j)^*\langle e_{f \circ \tau_{i+1}^j,1}, x_{f,j} \rangle &= (c_{j+1,i+1}^j \circ \sigma_{i+1}^j)^*\langle e_{f \circ \tau_b^j,1}, x_{f,j} \rangle \\
&= (\sigma_{(1,(j+1,i+1),(j))}^{j,1})^*\langle e_{f \circ \tau_b^j,1}, x_{f,j} \rangle \\
&= (\sigma_{i+1}^j)^*x_{f,j} \\
&= x_{f \circ \sigma_{i+1}^j,j} \\
&= (\sigma_{j+1}^j)^*e_{f \circ \sigma_{i+1}^j,1} \\
&= (\sigma_{\mathbf{j}+1,0}^j)^*\langle e_{f \circ \sigma_{i+1}^j,1} \rangle \\
&= (\rho_j \circ \tau_{j+1}^j)^*\langle e_{f \circ \sigma_{i+1}^j,1} \rangle \\
&= (\tau_{j+1}^j)^*(\rho_j)^*\langle e_{f \circ \sigma_{i+1}^j,1} \rangle
\end{aligned}$$

[Explain: $c_{j+1,i+1}^j \circ \sigma_{i+1}^j = \sigma_{(1,(j+1,i+1),(j))}^{j,1}$
Now, if $j < i$ we have

$$\begin{aligned}
&(\sigma_{i+1}^i)^*x_{f,j+1} = \\
&= (\sigma_{i+1}^i)^*(c_{i+1,j+1}^j)^*\langle (c_{j+1,i+1}^j)^*\langle e_{f \circ \tau_{i+1}^j,1}, x_{f,j} \rangle, (\rho_j)^*\langle e_{f \circ \sigma_{i+1}^j,1} \rangle \rangle \\
&= (c_{i+1,j+1}^j \circ \sigma_{i+1}^i)^*\langle (c_{j+1,i+1}^j)^*\langle e_{f \circ \tau_{i+1}^j,1}, x_{f,j} \rangle, (\rho_j)^*\langle e_{f \circ \sigma_{i+1}^j,1} \rangle \rangle \\
&= (\langle \sigma_{i+1}^i, \text{id}_{j+1} \rangle \circ c_{i,j+1}^j)^*\langle (c_{j+1,i+1}^j)^*\langle e_{f \circ \tau_{i+1}^j,1}, x_{f,j} \rangle, (\rho_j)^*\langle e_{f \circ \sigma_{i+1}^j,1} \rangle \rangle \\
&= (c_{i,j+1}^j)^*\langle (c_{j+1,i+1}^j \circ \sigma_{i+1}^i)^*\langle e_{f \circ \tau_{i+1}^j,1}, x_{f,j} \rangle, (\rho_j)^*\langle e_{f \circ \sigma_{i+1}^j,1} \rangle \rangle \\
&= (c_{i,j+1}^j)^*\langle (\langle \text{id}_{j+1}, \sigma_{i+1}^i \rangle \circ c_{j+1,i}^j)^*\langle e_{f \circ \tau_{i+1}^j,1}, x_{f,j} \rangle, (\rho_j)^*\langle e_{f \circ \sigma_{i+1}^j,1} \rangle \rangle \\
&= (c_{i,j+1}^j)^*\langle (c_{j+1,i}^j)^*\langle e_{f \circ \tau_{i+1}^j,1}, (\sigma_{i+1}^i)^*x_{f,j} \rangle, (\rho_j)^*\langle e_{f \circ \sigma_{i+1}^j,1} \rangle \rangle \\
&= (c_{i,j+1}^j)^*\langle (c_{j+1,i}^j)^*\langle e_{f \circ \tau_{i+1}^j,1}, x_{f \circ \sigma_{i+1}^i,j} \rangle, (\rho_j)^*\langle e_{f \circ \sigma_{i+1}^j,1} \rangle \rangle \\
&= (c_{i,j+1}^j)^*\langle (c_{j+1,i}^j)^*\langle e_{(f \circ \sigma_{i+1}^i) \circ \tau_i^j,1}, x_{f \circ \sigma_{i+1}^i,j} \rangle, (\rho_j)^*\langle e_{(f \circ \sigma_{i+1}^i) \circ \sigma_i^j,1} \rangle \rangle \\
&= x_{f \circ \sigma_{i+1}^i,j+1}
\end{aligned}$$

and

$$\begin{aligned}
& (\tau_{i+1}^i)^* x_{f,j+1} = \\
& = (\tau_{i+1}^i)^* (c_{i+1,j+1}^j)^* \langle (c_{j+1,i+1}^j)^* \langle e_{f \circ \tau_{i+1}^j, 1}, x_{f,j} \rangle, (\rho_j)^* \langle e_{f \circ \sigma_{i+1}^j, 1} \rangle \rangle \\
& = (c_{i+1,j+1}^j \circ \tau_{i+1}^i)^* \langle (c_{j+1,i+1}^j)^* \langle e_{f \circ \tau_{i+1}^j, 1}, x_{f,j} \rangle, (\rho_j)^* \langle e_{f \circ \sigma_{i+1}^j, 1} \rangle \rangle \\
& = (\langle \tau_{i+1}^i, \text{id}_{j+1} \rangle \circ c_{i,j+1}^j)^* \langle (c_{j+1,i+1}^j)^* \langle e_{f \circ \tau_{i+1}^j, 1}, x_{f,j} \rangle, (\rho_j)^* \langle e_{f \circ \sigma_{i+1}^j, 1} \rangle \rangle \\
& = (c_{i,j+1}^j)^* \langle (c_{j+1,i+1}^j \circ \tau_{i+1}^i)^* \langle e_{f \circ \tau_{i+1}^j, 1}, x_{f,j} \rangle, (\rho_j)^* \langle e_{f \circ \sigma_{i+1}^j, 1} \rangle \rangle \\
& = (c_{i,j+1}^j)^* \langle (\langle \text{id}_{j+1}, \tau_{i+1}^i \rangle \circ c_{j+1,i}^j)^* \langle e_{f \circ \tau_{i+1}^j, 1}, x_{f,j} \rangle, (\rho_j)^* \langle e_{f \circ \sigma_{i+1}^j, 1} \rangle \rangle \\
& = (c_{i,j+1}^j)^* \langle (c_{j+1,i}^j)^* \langle e_{f \circ \tau_{i+1}^j, 1}, (\tau_{i+1}^i)^* x_{f,j} \rangle, (\rho_j)^* \langle e_{f \circ \sigma_{i+1}^j, 1} \rangle \rangle \\
& = (c_{i,j+1}^j)^* \langle (c_{j+1,i}^j)^* \langle e_{f \circ \tau_{i+1}^j, 1}, x_{f \circ \tau_{i+1}^i, j} \rangle, (\rho_j)^* \langle e_{f \circ \sigma_{i+1}^j, 1} \rangle \rangle \\
& = (c_{i,j+1}^j)^* \langle (c_{j+1,i}^j)^* \langle e_{(f \circ \tau_{i+1}^i) \circ \tau_{i+1}^j, 1}, x_{f \circ \tau_{i+1}^i, j} \rangle, (\rho_j)^* \langle e_{(f \circ \tau_{i+1}^i) \circ \sigma_{i+1}^j, 1} \rangle \rangle \\
& = x_{f \circ \tau_{i+1}^i, j+1}
\end{aligned}$$

. (We can replace $\tau_{i+1}^i, \sigma_{i+1}^i$ with $\tau_{i+1}^a, \sigma_{i+1}^a$ for $j \leq a \leq i$ to get the required formulas.)

Now we compute

$$\begin{aligned}
(\sigma_{i+1}^i)^* x_{f,i+1} & = (\sigma_{i+1}^i)^* (c_{i+1,i+1}^i)^* \langle (c_{i+1,i+1}^i)^* \langle e_{f \circ \tau_{i+1}^i, 1}, x_{f,i} \rangle, (\rho_i)^* \langle e_{f \circ \sigma_{i+1}^i, 1} \rangle \rangle \\
& = (\sigma_{(1,(i+1,i+1),(i))}^{i,1})^* \langle (c_{i+1,i+1}^i)^* \langle e_{f \circ \tau_{i+1}^i, 1}, x_{f,i} \rangle, (\rho_i)^* \langle e_{f \circ \sigma_{i+1}^i, 1} \rangle \rangle \\
& = (\sigma_{i+1}^i)^* (\rho_i)^* \langle e_{f \circ \sigma_{i+1}^i, 1} \rangle \\
& = (\rho_i \circ \sigma_{i+1}^i)^* \langle e_{f \circ \sigma_{i+1}^i, 1} \rangle \\
& = (\tau_{\mathbf{i}+1,0}^i)^* \langle e_{f \circ \sigma_{i+1}^i, 1} \rangle \\
& = (\tau_{i+1}^i)^* e_{f \circ \sigma_{i+1}^i, 1} \\
& = e_{f \circ \sigma_{i+1}^i, 0} \\
& = (\sigma_{i+1}^i)^* e_{f,0}
\end{aligned}$$

and

$$\begin{aligned}
(\tau_{i+1}^i)^* x_{f,i+1} &= (\tau_{i+1}^i)^* (c_{i+1,i+1}^i)^* \langle (c_{i+1,i+1}^i)^* \langle e_{f \circ \tau_{i+1}^i, 1}, x_{f,i} \rangle, (\rho_i)^* \langle e_{f \circ \sigma_{i+1}^i, 1} \rangle \rangle \\
&= (\tau_{(1,(i+1,i+1),(i))}^{i,0})^* \langle (c_{i+1,i+1}^i)^* \langle e_{f \circ \tau_{i+1}^i, 1}, x_{f,i} \rangle, (\rho_i)^* \langle e_{f \circ \sigma_{i+1}^i, 1} \rangle \rangle \\
&= (\tau_{i+1}^i)^* (c_{i+1,i+1}^i)^* \langle e_{f \circ \tau_{i+1}^i, 1}, x_{f,i} \rangle \\
&= (c_{i+1,i+1}^i \circ \tau_{i+1}^i)^* \langle e_{f \circ \tau_{i+1}^i, 1}, x_{f,i} \rangle \\
&= (\tau_{(1,(i+1,i+1),(i))}^{i,0})^* \langle e_{f \circ \tau_{i+1}^i, 1}, x_{f,i} \rangle \\
&= (\tau_{i+1}^i)^* e_{f \circ \tau_{i+1}^i, 1} \\
&= e_{f \circ \tau_{i+1}^i, 0} \\
&= (\tau_{j+1}^j)^* e_{f,0},
\end{aligned}$$

which shows that $e_{f,1}$ is well-attached.

There are three “relative” variants of this cylinder construction that we will need.

5 Frustum Cylinder Functor

We prove functoriality of the frustum cylinder in the comma category of relative cell complexes.

This is done in a way compatible with the next section.

6 Ara Coherators

We prove that every Ara coherator is a coherator.

7 Model Categories of Models over a Coherator

We prove that the Grothendieck–Quillen model structure exists on the category of models over a coherator.

We prove that a cellular extension of coherators induces a Quillen equivalence on Grothendieck–Quillen model structures.

8 Homotopy Hypothesis

We prove that a natural transformation of topological realization induces a natural weak equivalence between the associated Kan extensions to models.

We prove that a topological realization induces a Quillen equivalence between models over a coherator and spaces and homotopies between topological realization give rise to natural weak equivalences between the corresponding Quillen adjunctions.